

MD Simulation Report – Myoglobin (1MBN)

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<https://github.com/wasitkrish/Material-Informatics>

Property	Value
Protein Name	Myoglobin
Protein ID	1MBN
Number of Chains	1
Number of Amino Acids	153
Net Charge	2
Total Atoms	31933
RMSD Avg	0.1709 nm
RMSD Std	0.0272 nm
RMSF Avg	0.0705 nm
RMSF Std	0.0237 nm
Radius of Gyration Avg	1.5447 nm
Radius of Gyration Std	0.0071 nm
Hydrogen Bonds Avg	122.0987 count
Hydrogen Bonds Std	5.5044 count
Pressure Avg	-17.4503 bar
Pressure Std	167.9087 bar
Temperature Avg	294.2119 K
Temperature Std	41.6338 K
Potential Energy Avg	-487078.0293 kJ/mol
Potential Energy Std	38132.2716 kJ/mol
Kinetic Energy Avg	81178.0156 kJ/mol
Kinetic Energy Std	504.0031 kJ/mol
Total Energy Avg	-393937.7343 kJ/mol
Total Energy Std	965.4104 kJ/mol
SASA Avg	85.7631 nm ²
SASA Std	1.8155 nm ²
Density Avg	1021.1249 kg/m ³
Density Std	5.5905 kg/m ³